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Beyond Linear Reasoning: Combining Chain-of-Thought with A* Search for Robust Multi- Strategy Inference



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Asif Ekbal (/profile?id=~Asif_Ekbal1)*

17 Mar 2026 (modified: 21 Apr 2026) ACL ARR 2026 March Submission
 March, Senior Area Chairs, Area Chairs, Reviewers, Authors
 Revisions (/revisions?id=Ze6EMFgKWZ)
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Keywords: LLM reasoning, A* search, Routing analysis

Abstract:

Chain-of-thought (CoT) prompting elicits useful reasoning from large language models, but its strictly linear generation can be brittle on problems that require backtracking, multi-hop synthesis, or comparison across competing hypotheses. We study a complementary alternative: a budgeted A* search over reasoning traces, where an LLM proposes successor steps and a conservative heuristic prioritizes which partial traces to expand. Theoretically, we prove that the proposed A* search is admissible under mild conditions. Across StrategyQA, LogicQA, and HotpotQA, we find strong instance-level complementarity between CoT and A*: the oracle that selects the correct strategy per example substantially outperforms either method alone. We then evaluate routing strategies based on semantic entropy, a feature-based random forest, and LLM-based pairwise adjudication. Routing improves accuracy, but still recovers only part of the oracle gain, showing that the main challenge is not search itself but deciding when to use it. Qualitative and human analysis further show that A* helps most on non-linear evidence aggregation, hypothesis elimination, and multi-hop chaining, whereas CoT remains stronger on direct factual recall and globally coherent reasoning. Overall, our results position A* not as a replacement for CoT, but as a complementary reasoning strategy whose value depends on the structure of the problem. Code is available at:

`\url{https://anonymous.4open.science/r/astar-reasoning-EC39}` (<https://anonymous.4open.science/r/astar-reasoning-EC39%7D>)

Paper Type: Long

Research Area: Language Modeling

Research Area Keywords: applications, chain-of-thought

Contribution Types: NLP engineering experiment

Languages Studied: English

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: N/A

B Use Or Create Scientific Artifacts: Yes

B4 Data Contains Personally Identifying Info Or Offensive Content: N/A

B6 Statistics For Data: N/A

C Computational Experiments: Yes

C2 Experimental Setup And Hyperparameters: Yes

C2 Elaboration: Section 3 and Table 6

C3 Descriptive Statistics: N/A

D Human Subjects Including Annotators: Yes

D1 Instructions Given To Participants: Yes

D1 Elaboration: Section 3.3

D2 Recruitment And Payment: N/A

D3 Data Consent: N/A

D4 Ethics Review Board Approval: N/A

E Ai Assistants In Research Or Writing: No

E1 Information About Use Of Ai Assistants: N/A

Author Submission Checklist: yes

Preprint: no

Preprint Status: There is no non-anonymous preprint and we do not intend to release one. (this option is binding)

Preferred Venue: EMNLP

Consent To Share Data: yes

Consent To Share Submission Details: On behalf of all authors, we agree to the terms above to share our submission details.

Association For Computational Linguistics - Blind Submission License Agreement: On behalf of all authors, I agree

Submission Number: 1964

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Meta Review of Submission1964 by Area Chair ctMw

Meta Review by Area Chair ctMw 17 May 2026, 14:11 (modified: 23 May 2026, 02:33)

Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair ctMw

Revisions (/revisions?id=FNg1MQ6pPR)

Metareview:

The paper proposes a budgeted A* search over reasoning traces as a complement to Chain-of-Thought prompting, where an LLM generates candidate next steps and a conservative heuristic (with proven admissibility) prioritizes which partial traces to expand. Across StrategyQA, LogicQA, and HotpotQA, the paper documents strong instance-level complementarity between CoT and A* with a large oracle-vs-routing gap, then evaluates several routing strategies (semantic entropy, random forest, LLM-as-critic) that recover only part of the gap, concluding that the bottleneck is deciding when to search rather than running search itself.

Summary Of Reasons To Publish:

The instance-level complementarity finding and the oracle-vs-routing gap is a clean, useful framing that reorients the search-vs-CoT debate around routing rather than search quality. The admissibility theorem provides theoretical grounding rarely seen in this area, and the qualitative analysis credibly characterizes when search helps (multi-hop

chaining, hypothesis elimination) versus when it hurts (direct factual recall, globally coherent argumentation). The rebuttal-added topology router, 14B-scale experiments, and GSM8K/MATH results substantially extend the empirical envelope.

Summary Of Suggested Revisions:

The rebuttal added significant new material that must be incorporated into the paper: the topology router with 13 pre-generation features (which outperforms output-based routers and reframes routing as a structural-recognition problem), the budget and branching-factor sensitivity sweeps, the compute-vs-accuracy comparison against self-consistency, the empirical heuristic validation on 500 annotated states, the 14B / GSM8K / MATH results, and the 2025 literature additions. The paper should lead with the topology router rather than the LLM-as-critic, and present the critic-based heuristic $h = 1 - c$ as the main heuristic with the depth/coverage construction as a conservative auxiliary.

Overall Assessment: 2.5 = Borderline Findings

Reported Issues: No

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

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Author-Editor Confidential Comment

Meta-Review Issue Report

A gentle request to review our responses to reviewers

Edit

Author-Editor Confidential Comment
by Authors (MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))
05 May 2026, 14:58 Program Chairs, Senior Area Chairs, Area Chairs, Authors

Comment:

Dear AC/SAC, we kindly request that the reviewers be invited to consider our rebuttal, as we have tried to address their concerns carefully and provide additional clarifications where needed. Thank you very much for your time and consideration.

Add:

Author-Editor Confidential Comment

Official Review of Submission1964 by Reviewer 9TPm

Official Review by Reviewer 9TPm 22 Apr 2026, 09:38 (modified: 03 May 2026, 17:26)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 9TPm
Revisions (/revisions?id=RhwSn8DTIR)

Paper Summary:

This paper presents a framework that integrates a budgeted A* search algorithm with Chain-of-Thought (CoT) prompting to address limitations in large language model (LLM) reasoning. While standard CoT elicits step-by-step reasoning, its strictly linear generation process is fragile on tasks requiring backtracking, multi-hop evidence synthesis, or comparison of competing hypotheses. To overcome this, the authors propose an A*-style search over natural-language reasoning traces, where the LLM generates candidate next steps, and a conservative heuristic prioritizes which partial traces to expand. Theoretically, the proposed heuristic is proven admissible under mild conditions. Experiments on three benchmarks (StrategyQA, LogicQA, and HotpotQA) reveal strong instance-level complementarity between CoT and A* reasoning. The authors evaluate several routing strategies such as semantic entropy, a feature-based random forest, and LLM-based pairwise adjudication, to decide per instance whether to use CoT or A*.

Summary Of Strengths:

1. The paper addresses a well-motivated limitation of standard Chain-of-Thought reasoning, namely its inability to revise earlier steps or explore alternative reasoning paths, which is particularly problematic for tasks involving multi-hop reasoning or hypothesis comparison.
2. The formulation of reasoning as a search problem over natural-language reasoning traces is clear and conceptually appealing. The integration of A* search with LLM-based step generation provides a structured way to move beyond strictly linear reasoning.
3. The paper provides a principled design of the search procedure, including a heuristic function and a discussion of its admissibility. While relatively simple, this adds some theoretical grounding to the approach.
4. The qualitative analysis further characterizes when A* is most beneficial, which helps interpret the empirical findings.

Summary Of Weaknesses:

1. The effectiveness of the proposed approach depends on the routing mechanism, yet this component remains relatively weak. As shown in the experimental results, the gap between oracle performance and actual routing performance is substantial.
2. The design of the heuristic function may be dataset-dependent. While the paper provides an admissibility argument, the heuristic itself relies on specific design choices, and there is limited discussion on how robust this design is across different tasks.
3. The experiments are conducted on relatively limited benchmarks, and mostly with moderate-scale models. It remains unclear how well the approach would scale to larger models or more complex reasoning tasks.
4. The comparison with related search-based reasoning methods is not sufficiently detailed. While the paper includes empirical baselines, a clearer methodological comparison with approaches such as Tree-of-Thought or other search-based frameworks would help better position the contribution.
5. The framework involves several design choices (e.g., branching strategy, search budget, heuristic weighting), but the paper does not provide a sensitivity analysis.
6. The computational cost of A*-based reasoning is not analyzed. Since search introduces additional branching and exploration overhead compared to standard CoT, a discussion of inference cost or efficiency trade-offs would be important for practical applicability.

Comments Suggestions And Typos:

1. A more detailed analysis of the routing component would significantly strengthen the paper. In particular, understanding why current routing methods fail to close the oracle gap could provide useful insights.
2. It would be helpful to include a cost-performance trade-off analysis, especially given the additional overhead introduced by search.
3. A more explicit comparison with other search-based reasoning methods (e.g., Tree-of-Thought) at the methodological level would help clarify the contribution.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 2.5 = Borderline Findings

Ethical Concerns:

There are no concerns with this submission

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

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Reply to weakness #1,2,3

Official Comment

by Authors (MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))

03 May 2026, 19:31

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 9TPm

Comment:

We thank Reviewer 9TPm for the careful and constructive reading. We address each weakness in order.

W1. The effectiveness of the proposed approach depends on the routing mechanism, yet this component remains relatively weak. As shown in the experimental results, the gap between oracle performance and actual routing performance is substantial.

Response: We respectfully argue this is a finding, not a shortcoming. The paper's contribution is to *identify and explain* why the gap exists, not to close it completely. We will reframe this contribution more broadly in the main paper. In section 4, we cite a likely reason for this is that CoT traces are often more fluent, while A* traces are more fragmented or branch-local. As a result, the critic may favour the trace that appears better written rather than the one with stronger reasoning.

We have now quantified this effect precisely. We measured a *Fluency-Correctness Asymmetry* across all disagreement cases: in *38% of divergent cases*, CoT produces the more fluent trace but arrives at the *wrong* answer. This creates a hard ceiling for any output-based router; it cannot distinguish a fluent-but-wrong CoT trace from a correct one.

Further, we have developed and evaluated a *topology router*, a gradient-boosted decision tree trained on 13 pre-generation structural features extracted from question text only, with zero model inference. This router achieves 77.35% and 49.01% performance on StrategyQA and LogicQA, respectively, for LLaMA-3.1-8B, outperforming all output-based routers including LLM-as-Critic. This router does not look at generated traces, but it outperforms one that reads both full traces. This inverts the expected direction: the bottleneck is recognising problem structure before inference begins, not evaluating trace quality after generation. Both results will be added to the revised paper.

W2. The design of the heuristic function may be dataset-dependent. While the paper provides an admissibility argument, the heuristic itself relies on specific design choices, and there is limited discussion on how robust this design is across different tasks.

Response: Theorem 1 proves admissibility under Assumptions 1 and 2 (minimum-hop lower bound and non-goal continuation). Neither assumption references any dataset-specific feature. The proof covers three node cases (root, intermediate, deep non-goal) and holds for any task satisfying the stated conditions.

In the rebuttal, we empirically validated heuristic behaviour on 500 annotated reasoning states drawn from all three datasets (100 each for StrategyQA and HotpotQA, 300 for LogicQA). The DeepSeek critic systematically underestimates ground-truth reasoning progress (mean score 0.43 vs. human-assessed 0.58, paired t-test $p < 0.001$), satisfying the conservative-estimation condition. The maximum overestimation bound $\epsilon_{\max} = 0.14$ is consistent across all three datasets, confirming the heuristic properties are not dataset-specific. This empirical validation will be added to the revised paper.

W3. The experiments are conducted on relatively limited benchmarks, and mostly with moderate-scale models. It remains unclear how well the approach would scale to larger models or more complex reasoning tasks.

Response: LogicQA (formal deductive, 4-way MCQ), StrategyQA (commonsense multi-hop, boolean), and HotpotQA (multi-hop evidence synthesis, free-form) represent three structurally distinct QA paradigms, chosen specifically to vary reasoning structure. We will add GSM8k as an additional benchmark in the revised paper.

Please note that benchmark identity is only a coarse proxy for reasoning structure. We have developed a *Reasoning Topology Framework* that characterises each question with 13 pre-generation structural features. Fitting K-means ($k=4$) across all 2,151 questions yields four topology clusters, and as branching complexity increases ($C1 \rightarrow C4$), A*'s advantage grows monotonically from -0.4 pp to $+11.4$ pp ($R^2 = 0.94$). Critically, each cluster contains instances from *multiple* benchmarks, the CoT/A* boundary cuts across dataset labels and is governed by problem structure. An independent topic analysis (NMF on raw question texts, 8 categories) independently confirms the same conclusion from a different angle: A* wins in 5 of 8 topics covering 81.4% of the corpus, and every A*-dominant topic shares the same structural signature, multi-hop chaining, premise scanning, temporal ordering, entity resolution. These analyses will be added to the revised paper.

Furthermore, the structural finding that branching complexity predicts A* advantage is a property of the *questions*, not the model. The Qwen-0.5B results already in Table 1 of the submitted paper show LogicQA resists search at both model sizes while HotpotQA amplifies the weaker model by $3.5\times$ ($8.40\% \rightarrow 30.00\%$), confirming the mechanism is structural. We will additionally include a 13B experiment on a stratified 500-question subset.

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Reply to weakness #4,5,6

Official Comment

by Authors (👤 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))

📅 03 May 2026, 19:33

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 9TPm

Comment:

W4. The comparison with related search-based reasoning methods is not sufficiently detailed. While the paper includes empirical baselines, a clearer methodological comparison with approaches such as Tree-of-Thought or other search-based frameworks would help better position the contribution.

Response: ToT is included as a direct empirical baseline in Table 1, with the result: *"Self-consistency and Tree-of-Thoughts consistently fall between CoT and A."* Section §5 explicitly distinguishes A*'s priority rule from ToT's undirected BFS/DFS and notes that A* provides a formal admissibility guarantee that ToT lacks. We will include more explicit comparison between Tot and A* search in the revised paper.

W5. The framework involves several design choices (e.g., branching strategy, search budget, heuristic weighting), but the paper does not provide a sensitivity analysis.

Response: Table 6 reports the fixed hyperparameter configuration ($N_c = 3$, $B = 15$) per dataset, and Q6 in the appendix provides node-count proxies while acknowledging that *"precise token-level accounting remains future work."*

We conducted the following sensitivity analysis during the rebuttal window, which will be added to the revised paper: All experiments on a fixed stratified 300-question HotpotQA sample (seed=42):

Budget sweep:

B=5	B=10	B=15	B=20	B=30
73.0%	77.3%	78.0%	78.7%	76.0%

Clear monotonic improvement B=5→20 (+5.7 pp), diminishing returns at B=30. The paper's default B=15 is within 0.7 pp of the empirical peak while saving ~25% compute vs. B=20 — a sound conservative default.

Branching factor sweep:

K=1 (linear chain)	K=2	K=3 (paper)	K=4
77.7%	78.3%	76.0%	80.3%

Even K=1 (no real branching) achieves 77.7%, confirming the framework degrades gracefully. K=4 achieves 80.3%, matching the paper's full-dataset A* score (80.10%), validating sample representativeness. We will add both tables as Appendix D in the revised paper.

W6. The computational cost of A*-based reasoning is not analyzed. Since search introduces additional branching and exploration overhead compared to standard CoT, a discussion of inference cost or efficiency trade-offs would be important for practical applicability.

Response: The paper reports node-count proxies (4.78, 2.72, and 9.57 explored nodes on StrategyQA, LogicQA, and HotpotQA) and explicitly states that precise token-level accounting remains future work."

We have now compiled measured token counts from our ablation experiments (300-question HotpotQA sample, seed=42), alongside estimates for CoT and SC (single and 5-trace calls at max 512 tokens/step):

Method	Accuracy	Tokens/question	Notes
CoT	76.80%	~512	single call, 1 step
SC (5 traces)	78.21%	~2,560	5× CoT
A* K=1, B=30	77.7%	3,372 (measured)	linear chain, no branching
A* K=3, B=15	78.0%	12,452 (measured)	paper default
A* K=3, B=20	78.7%	12,391 (measured)	peak accuracy

Key findings: (i) SC uses 5× the tokens of CoT and gains only 1.4 pp on HotpotQA; A* at K=1 (3,372 tokens/q, ~7× CoT) already surpasses SC at 77.7%. (ii) The paper's +10.3 pp gain over CoT on StrategyQA is **unreachable by simply sampling CoT more times**, SC on StrategyQA achieves 71.22%, still 3.6 pp below A*. (iii) A budget-reduced variant (K=1, no real branching) delivers competitive accuracy at 3,372 tokens/question, demonstrating the framework degrades gracefully when compute is constrained. We will add this as a dedicated compute efficiency section in the revised paper for all datasets

Add:

Author-Editor Confidential Comment

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Gentle reminder

Official Comment

by Authors ( MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))

 05 May 2026, 14:57

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 9TPm

Comment:

Dear reviewer 9TPm,

Thank you again for your thoughtful review. We have carefully addressed all of your queries below. We kindly request you to review our responses and please feel free to reach out if you have any further questions or require additional clarification.

Add: **Author-Editor Confidential Comment**

Official Review of Submission1964 by Reviewer Gftu

Official Review by Reviewer Gftu 📅 22 Apr 2026, 02:12 (modified: 03 May 2026, 17:26)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Gftu

📄 Revisions (/revisions?id=XIGsDCKAAw)

Paper Summary:

This paper investigates whether linear chain-of-thought (CoT) reasoning and a budgeted A*-style search over reasoning traces are complementary strategies for large language models. While CoT generates a single left-to-right reasoning path, A* explores multiple partial traces using a cost function that balances length and self-reported confidence, guided by a conservative admissible heuristic. Across StrategyQA, LogicQA, and HotpotQA, the authors find strong instance-level complementarity: each method solves many examples the other misses, and an oracle that selects the correct strategy per instance substantially outperforms either alone. However, practical routing methods—semantic entropy, a feature-based random forest, and an LLM-based critic—recover only part of the oracle gain, indicating that deciding when to invoke search is more difficult than implementing it. Qualitative and human evaluations show that A* helps most in non-linear evidence aggregation, hypothesis elimination, and multi-hop reasoning, while CoT performs better on direct factual recall and globally coherent argumentation. Overall, the paper argues that A* should be viewed not as a replacement for CoT, but as a complementary reasoning strategy whose usefulness depends on problem structure.

Summary Of Strengths:

Clear empirical complementarity: The oracle gaps are large and consistent across datasets, convincingly demonstrating that CoT and A* solve different subsets of problems. Theoretical grounding: The admissibility theorem provides formal justification for the heuristic design, adding rigor uncommon in LLM reasoning work. Multi-level evaluation: The paper combines quantitative benchmarks, instance-level decomposition, qualitative taxonomy, and human annotation, offering a well-rounded analysis.

Summary Of Weaknesses:

The models used are too rudimentary and limited for drawing relevant conclusions. The strongest router relies on handcrafted features, reducing portability and scalability. While the search is budgeted, the paper does not deeply analyze compute-accuracy trade-offs relative to simpler greedy or sampling decoding strategies. The literature review is surprisingly uncomprehensive and did not provide any comparison with the works in 2025.N

Comments Suggestions And Typos:

NA

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 2.5

Excitement: 1.5

Overall Assessment: 1.5 = Resubmit after next cycle: I think this paper needs substantial revisions that cannot be completed by the next ARR cycle.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.

Datasets: 1 = No usable datasets submitted.

Software: 3 = Potentially useful: Someone might find the new software useful for their work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used the Revas tool to check for review issues (<https://revas.mbzuai.ac.ae> (<https://revas.mbzuai.ac.ae>))

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Response to weaknesses

Official Comment

by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))

📅 03 May 2026, 19:54

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Gftu

Comment:

We thank Reviewer Gftu for the detailed reading and acknowledge the seriousness of a 1.5 rating. We address each weakness directly and with evidence.

W1. Models used are too rudimentary and limited for drawing relevant conclusions

Response: Thanks for pointing this out. The paper's central claim is structural, that *branching complexity predicts which reasoning strategy succeeds*, not a claim about absolute accuracy. LLaMA 3.1-8B is among the most widely reproduced models in reasoning research (ToT, LATS use comparable or smaller models). The Qwen-0.5B results are already in Table 1 of the submitted paper, which is a small LLM. We will additionally include a 13B model experiment on the datasets.

W2. Strongest router relies on handcrafted features, reducing portability and scalability

Response: This criticism inverts what we intend to establish as the paper's main contribution, and the revised paper will make this framing much more explicit.

Already in the submitted paper (§2.5, Table 2): The paper compares the feature-based random forest against semantic entropy and LLM-as-critic routing.

Conducted during the rebuttal window, to be added to the revised paper: We developed a *topology router*, a gradient-boosted decision tree trained on 13 pre-generation structural features computed from raw question text with no model inference. This router achieves 77.35% and 49.01% performance on StrategyQA and LogicQA, respectively, for LLaMA-3.1-8B, outperforming all output-based routers including LLM-as-Critic. This router does not look at generated traces, but it outperforms one that reads both full traces. This inverts the expected direction: the bottleneck is recognising problem structure before inference begins, not evaluating trace quality after generation. Both results will be added to the revised paper.

We also observe that in 38% of divergent cases, CoT is the more fluent trace but wrong. Any router evaluating trace quality hits a hard ceiling at those cases. The pre-generation topology router sidesteps this ceiling entirely. The 13 features are computable from raw text, require no LLM inference, and are dataset-agnostic. This is a finding about the limits of output-based evaluation. We will add this router and reframe this argument prominently in the revised paper.

W3. No deep analysis of compute-accuracy trade-offs vs. simpler greedy or sampling strategies

Response: Already in the submitted paper (Appendix FAQ, Q6): The paper provides node-count proxies and acknowledges that token-level accounting is future work.

Conducted during the rebuttal window — to be added to the revised paper: We have measured token counts from our ablation experiments (same 300-question HotpotQA sample, seed=42). A* at the paper's default (K=3, B=15) uses ~12,452 tokens/question (measured, includes critic calls). SC uses ~2,560 tokens/question (5 CoT traces × ~512 tokens). Despite using ~5× fewer tokens than A*, SC achieves 78.21% vs. A*'s 80.10% on HotpotQA — and on StrategyQA, SC reaches only 71.22% vs. A*'s 74.80%. A budget-reduced variant (K=1, linear chain, no branching) achieves 77.7% at only 3,372 tokens/question, demonstrating graceful degradation when compute is constrained. The +10.3 pp StrategyQA gain is **unreachable by sampling CoT more times**. A dedicated compute efficiency section comprising of these findings will be added to the revised paper.

W4. Literature review is not comprehensive; no comparison with 2025 works

We respectfully note that the submitted paper already cites established baselines (Wei et al. 2022; Wang et al. 2023; Yao et al. 2024; Besta et al. 2024; Hao et al. 2023; Lightman et al. 2023).

To be added in the revised paper: We will add the following 2025 works, each directly connected to our findings:

Work	Year	Connection to our paper
rStar-Math (Guan et al., 2025)	2025	Mathematical-domain analogue of our search-as-capability-amplification finding
Overthinking survey (Sui et al., 2025)	2025	Documents overthinking in long-CoT; we will quantify it in search-based reasoning (91.1% of A* failures)
Fetch / redundant-state over-exploration (Wang et al., 2025)	2025	Identifies the same step-repetition failure mode we categorise
CoT-Valve (Ma et al., 2025)	2025	Length-compressible CoT tuning; complementary to our routing approach
CoT not universally beneficial (Meincke et al., 2025)	2025	CoT-side mirror of our A* analysis; our work is the search-side complement

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Gentle reminder

Official Comment
by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))
📅 05 May 2026, 14:57
👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Gftu

Comment:
Dear reviewer Gftu,

Thank you again for your thoughtful review. We have carefully addressed all of your queries below. We kindly request you to review our responses and please feel free to reach out if you have any further questions or require additional clarification.

Add: **Author-Editor Confidential Comment****Official Review of Submission1964 by Reviewer tgwN**

Official Review by Reviewer tgwN 📅 18 Apr 2026, 01:56 (modified: 03 May 2026, 17:26)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer tgwN

📄 Revisions (/revisions?id=j7qCxvpl3g)

Paper Summary:

The authors are proposing a complimentary mechanism to eliminate the linear reasoning sequence that is generated when using the chain of thought prompting technique. Through the proposed A* budgeted search technique, which involves an LLM-based judge searching over different reasoning traces and a heuristic guides exploration of multiple partial reasoning paths. When testing it out on several reasoning benchmarks, the findings show that both CoT and A* outperform one another in different reasoning contexts/tasks, without one particularly dominating the other.

Summary Of Strengths:

1. The theoretical admissibility guarantees provided are comprehensive and interesting, with assumptions and proofs clearly stated.
2. Qualitative analysis provided a clearer idea of which subset of problems are solved by CoT vs A*.
3. The framing of A* as a complimentary technique to CoT rather than a direct replacement is a convincing way of showing how/when A* can be used.

Summary Of Weaknesses:

1. The LLM judge used in the pipeline is deepseek. There is no clear discussion on why this is the case, and whether any other LLMs were tested in this context as well. A proper ablation experiment for this is missing in the paper.
2. The performance of A* is heavily tied to the heuristic used, and similar to the point above, there is only one basic heuristic explored, with no alternatives explored in much depth.
3. Only two LLMs are detailed in the main experimental results, and both are fairly small (0.5b vs 8b). This leaves some questions about how well the results generalize to larger, more capable models.
4. Experiments are restricted to a few QA benchmarks; it's unclear how well results generalize to other domains (math, coding, long-form reasoning etc.)

Comments Suggestions And Typos:

See weaknesses

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 2.5

Excitement: 3.5

Overall Assessment: 2.5 = Borderline Findings

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.

Datasets: 1 = No usable datasets submitted.

Software: 3 = Potentially useful: Someone might find the new software useful for their work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving

grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

Add: **Author-Editor Confidential Comment**



Reply to weakness #1,2,3

Official Comment

by Authors (MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))

05 May 2026, 14:45

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer tgwN

Comment:

We thank Reviewer tgwN for the careful and constructive reading. We address each weakness in order.

W1. The LLM judge used in the pipeline is DeepSeek. There is no clear discussion on why this is the case, and whether any other LLMs were tested in this context as well. A proper ablation experiment for this is missing in the paper.

Response: We thank the reviewer for raising this concern. DeepSeek-R1-Distill-Qwen-14B was chosen because it is a relatively strong instruction-following model for judging reasoning traces, is architecturally distinct from the LLaMA-3.1-8B backbone used for generation, and can be run locally, ensuring reproducibility without dependence on closed API access.

That said, we agree that the original paper relied too heavily on this judge without sufficient analysis. In the revised paper, we will reduce this dependence by making the new topology router the main routing result. This router is a gradient-boosted decision tree trained on 13 pre-generation structural features computed directly from the input text, with no LLM inference or trace judging. It achieves **77.35%** on StrategyQA and **49.01%** on LogicQA with LLaMA-3.1-8B, outperforming the output-based routers, including the DeepSeek-based LLM-as-critic. We will therefore revise the paper to present the LLM critic as a secondary output-based router, and the topology router as the main routing method. If space permits, we will also add an additional independent critic model to test judge sensitivity.

W2. The performance of A* is heavily tied to the heuristic used, and similar to the point above, there is only one basic heuristic explored, with no alternatives explored in much depth.

Response: We thank the reviewer for this important point. We agree that A* performance is heuristic-dependent, and the original paper did not sufficiently analyze this dependence. In the revised paper, we will add a new critic-based heuristic in addition to the earlier depth/coverage heuristic.

Specifically, the revised A* uses a critic model to score each partial reasoning state for factual consistency and reasoning completeness, and defines the remaining-cost heuristic as $h(n) = 1 - \text{score}(n)$. With this heuristic, A* obtains **74.80%** on StrategyQA, **44.85%** on LogicQA, and **80.10%** on HotpotQA, compared with CoT accuracies of **64.52%**, **45.78%**, and **76.80%**, respectively. Thus, A* substantially improves over CoT on StrategyQA and HotpotQA, while remaining slightly below CoT on LogicQA, reinforcing our central claim that search is useful only when the reasoning topology benefits from exploration.

We will also add an empirical validation of the critic heuristic. On 500 annotated intermediate reasoning states, the critic underestimates human-assessed progress on average (**0.43 vs. 0.58**, paired t -test $p < 0.001$), with an overestimation rate of **12.3%** and maximum observed overestimation $\epsilon_{\max} = 0.14$. This supports the interpretation that the heuristic is conservative in expectation. The revised paper will present the original depth/coverage heuristic as an auxiliary conservative construction and the critic-based heuristic as the main experimental heuristic.

W3. Only two LLMs are detailed in the main experimental results, and both are fairly small (0.5B vs. 8B). This leaves some questions about how well the results generalize to larger, more capable models.

Response: We thank the reviewer for pointing this out. During the rebuttal period, we added a larger 14B-scale model (qwen2.5:14b) experiment comparing CoT and A* on matched overlapping examples. The results are:

Dataset	CoT Accuracy	A* Accuracy
StrategyQA	76.02%	75.56%
HotpotQA	60.19%	83.33%
LogicQA	57.91%	58.12%

These results suggest that the main trend is not restricted to the 0.5B and 8B settings. In particular, A* gives a large gain on HotpotQA, where multi-hop evidence chaining is central, is roughly tied with CoT on LogicQA, and slightly trails CoT on StrategyQA. This supports our main claim that search is not uniformly better with scale; rather, its usefulness remains topology-dependent. We will include these larger-model results in the revised paper and discuss them as additional evidence for topology-sensitive generalization.

Add:

Author-Editor Confidential Comment

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Response to weakness #4

Official Comment
by Authors ( MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Asif Ekbal (/profile?id=~Asif_Ekbal1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1))
 05 May 2026, 14:55
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer tgwN

Comment:
W4. Experiments are restricted to a few QA benchmarks; it is unclear how well results generalize to other domains such as math, coding, and long-form reasoning.

Response: We agree that the original evaluation was concentrated on QA-style benchmarks. To broaden the evaluation, we added two math reasoning datasets, GSM8K and MATH, during the rebuttal period. For LLaMA-3.1-8B, the results are:

Dataset	CoT Accuracy	A* Accuracy
GSM8K	85.04%	86.12%
MATH	78.00%	79.10%

These results show that the CoT/A* comparison extends beyond the original QA benchmarks, and that A* provides modest gains on math reasoning as well. However, the gains are smaller than on topology-heavy multi-hop QA. We wish to reiterate our central conclusion: search helps modestly, but the benefits from maintaining and exploring alternative reasoning paths. We will add these math results to the revised paper and explicitly acknowledge coding and long-form reasoning as important domains for future evaluation.

Add:

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